

Weld-Bonding



General Equipment

3 Phase Inverter Spot Welder 254-00002
Compuspot 700F Welder 190-50080
I4 Inverter Spot Welder 254-00014
Inverter Welder with MIG Welder 254-00015

Material

Item	Specification
Motorcraft® Metal Bonding Adhesive TA-1	—
Motorcraft® Metal Surface Prep ZC-31-B	—
Rust Inhibitor ValuGard™ VG104, VG104A (aerosol)	—
Motorcraft® Seam Sealer TA-2	—

Weld-Bonding — Squeeze-Type Resistance Spot Welding (STRW) Method

WARNING: Invisible ultraviolet and infrared rays emitted in welding can injure unprotected eyes and skin. Always use protection such as a welder's helmet with dark-colored filter lenses of the correct density. Electric welding will produce intense radiation, therefore, filter plate lenses of the deepest shade providing adequate visibility are recommended. It is strongly recommended that persons working in the weld area wear flash safety goggles. Also wear protective clothing. Failure to follow these instructions may result in serious personal injury.

WARNING: Always refer to Material Safety Data Sheet (MSDS) when handling chemicals and wear protective equipment as directed. Examples may include but are not limited to respirators and chemically resistant gloves. Failure to follow these instructions may result in serious personal injury.

WARNING: Always wear protective equipment including eye protection with side shields, and a dust mask when sanding or grinding. Failure to follow these instructions may result in serious personal injury.

WARNING: Collision damage repair must conform to the instructions contained in this workshop manual. Replacement components must be new, genuine Ford Motor Company parts. Recycled, salvaged, aftermarket or reconditioned parts (including body parts, wheels or safety restraint components) are not authorized by Ford.

Departure from the instructions provided in this manual, including alternate repair methods or the use of substitute components, risks compromising crash safety. Failure to follow these instructions may adversely affect structural integrity and crash safety performance, which could result in serious personal injury to vehicle occupants in a crash.

NOTE: On door shells that are manufactured with structural adhesives only, weld bonding door skins is not recommended. Only metal bonding adhesive should be used.

NOTE: Weld-bonding is a method used to join metals using Squeeze-Type Resistance Spot Welding (STRW) or Metal Inert Gas (MIG) welding and structural adhesive. The steps listed in this procedure apply to both types of welding. STRW is the preferred method. MIG welding should only be used when areas to be welded cannot be accessed using STRW-type machinery.

NOTE: Factory spot welds should be substituted with either resistance spot welds or MIG plug welds. Spot/plug welds should equal factory welds in both location and quantity. Do not place a new spot weld directly over an original weld location. Plug weld hole should equal 8 mm (0.31 in) diameter.

1. Verify the vehicle is dimensionally correct on a frame machine. Straighten if necessary.
2. Remove damaged panels with an air saw or air chisel. Remove only large portions of the damaged panel. Avoid cutting into mating flanges or adjacent parts.
3. Drill out the spot welds using an appropriate spot-weld cutter and remove the remaining portions of the panel to be replaced.
4. Prepare any damaged flanges on the vehicle using hammer and dolly.
5. Grind the mating surface of the original flanges no greater than 25 mm (0.98 in) where the metal bonding adhesive will be applied.
 - Be sure to remove galvanizing on metal. Metal should have a shiny appearance.
 - Be careful not to damage the corners or thin the metal. The E-coat should also be removed on the opposite side of the flange only where the spot welds are to be placed. Clean surfaces with metal surface prep or equivalent.
6. Dry-fit and clamp the replacement service parts to verify a correct fit.
 - Remove the service part after verifying correct fit and alignment.
7. **NOTE:** The ends of welding clamps should be insulated on the ends using tape or similar material when welding is carried out.

Follow manufacturer's prescribed welding procedures and settings. For additional information, refer to Welding Precautions — Steel in this section.

8. Prepare the adhesive. Dispense a small amount of metal bonding adhesive from the cartridge to make sure of an even flow of both components. Attach the mixing tip and dispense a mixing tip length of adhesive to make sure of a correct mix ratio.
9. **NOTE:** Welding can be carried out anytime during the adhesive curing process, or after the adhesive is fully cured. Welder settings will vary when welding through wet adhesive versus welding through fully cured adhesive. Refer to welder manufacturer's recommended settings for welding through fully cured adhesive. It is recommended to place a shunt weld in an area with no adhesive to make sure of conductivity, particularly when welding through fully cured adhesive.

NOTE: Refer to the product label for handling and preparation instructions.

Create a test sample.

1. Prepare the metal and adhesive as described. Apply a 6-9 mm (0.23-0.35 in) bead of adhesive and weld the sample.
 2. Place the welded sample in a vice and carry out destructive weld tests by peeling the scrap metal apart using large lock-type pliers. Measure the weld nugget to determine that the nugget meets Ford weld nugget requirements. If the weld nugget does not meet required size, adjust welder settings until the correct weld nugget size is achieved.
 3. When the correct weld nugget size is achieved, the service part can be weld-bonded. For additional information, refer to the weld nugget table in Specifications in this section.
10. Apply a 6-9 mm (0.23-0.35 in) bead of metal bonding adhesive to the vehicle prepared flange surface.
 11. Place the service part(s) in the correct position on the vehicle.

- When positioned, do not pull the component away from the vehicle. If repositioning is necessary, slide the service part(s). This will make sure of correct contact between the components and adhesive.

12. Clamp evenly and tightly. The adhesive contains glass beads which will prevent over-clamping the component.

13. **NOTE:** *Refer to product label for cure times and handling instructions. Clamps may be removed immediately after the component is welded.*

Wipe excess adhesive from the panel before it cures.

14. Finish any cosmetic section seams with fiber-filled body filler. Rough sand the filler, apply conventional body filler after the adhesive cures and block-sand the area.

15. Use seam sealer wherever a cosmetic seam sealer is required.

16. Mix and apply primer surfacer per Ford-approved paint recommendations.

17. Mix and apply basecoat per Ford-approved paint recommendations.

18. Mix and apply clearcoat per Ford-approved paint recommendations. Refinishing materials may be force-dried following paint manufacturer's recommendations.

19. Apply anti-corrosion treatment to the repair area as required. For additional information, refer to Restoring Corrosion Protection Following Repair in this section.

Weld-Bonding — Metal Inert Gas (MIG) Welding Method

WARNING: Collision damage repair must conform to the instructions contained in this workshop manual. Replacement components must be new, genuine Ford Motor Company parts. Recycled, salvaged, aftermarket or reconditioned parts (including body parts, wheels or safety restraint components) are not authorized by Ford.

Departure from the instructions provided in this manual, including alternate repair methods or the use of substitute components, risks compromising crash safety. Failure to follow these instructions may adversely affect structural integrity and crash safety performance, which could result in serious personal injury to vehicle occupants in a crash.

WARNING: Invisible ultraviolet and infrared rays emitted in welding can injure unprotected eyes and skin. Always use protection such as a welder's helmet with dark-colored filter lenses of the correct density. Electric welding will produce intense radiation, therefore, filter plate lenses of the deepest shade providing adequate visibility are recommended. It is strongly recommended that persons working in the weld area wear flash safety goggles. Also wear protective clothing. Failure to follow these instructions may result in serious personal injury.

WARNING: Always refer to Material Safety Data Sheet (MSDS) when handling chemicals and wear protective equipment as directed. Examples may include but are not limited to respirators and chemically resistant gloves. Failure to follow these instructions may result in serious personal injury.

WARNING: Always wear protective equipment including eye protection with side shields, and a dust mask when sanding or grinding. Failure to follow these instructions may result in serious personal injury.

NOTE: *Metal Inert Gas (MIG) weld-bonding can be substituted as an alternative to Squeeze-Type Resistance Spot Welding (STRW) . It may only be used to weld areas that are inaccessible to STRW machinery. However, when accessible, STRW is the preferred method.*

NOTE: *On door shells that are manufactured with structural adhesives only, weld-bonding door skins is not recommended. Only metal bonding adhesive should be used.*

NOTE: *Factory spot welds should be substituted with either resistance spot welds or MIG plug welds. Spot/plug welds should equal factory welds in both location and quantity. Do not place a new spot weld directly over an original weld location. Plug weld hole should equal 8 mm (0.31 in) diameter.*

1. Remove damaged panels with an air saw or air chisel. Remove only large portions of the damaged panel. Avoid cutting into mating flanges or adjacent parts.
 - Drill out the spot welds using an appropriate spot-weld cutter and remove the remaining portions of the panel to be replaced.
2. After removing the damaged sheet metal panel(s), repair any damaged flanges on the vehicle using a hammer and dolly.
3. Using an appropriate grinder, carefully grind around the entire receiving flange area following the original welds. Be sure to remove all E-coat, paint or galvanized coating from the mating surfaces of the joint.
 - Be sure to remove galvanizing on metal. Metal should have a shiny appearance.
 - Be careful not to damage the corners or thin the metal. The E-coat should also be removed on the opposite side of the flange only where the spot welds are to be placed. Clean the surface with metal surface prep or equivalent.
4. Repeat the procedure from Step 3 on the mating surface of the replacement service part(s).
5. Prepare the new service panel for plug welds.
 - Using the original panel as a reference, drill or punch 8 mm (0.31 in) diameter holes in the exact number as the original spot welds. The holes should be positioned as close as possible to the original spot weld locations, without lining up exactly on top of an original spot weld site.
 - To make sure of correct weld performance, grind the immediate perimeter of the plug weld hole. Grind only in the area of the plug weld; this will keep corrosion to a minimum.
6. Dry-fit and clamp the replacement service parts to verify a correct fit and alignment.
 - Remove the service part after verifying correct fit and alignment.
7. The vehicle prepared flange areas where plug welds will be located must be kept free of adhesive. Apply 25 mm (0.98 in) tape to the plug weld areas to prevent contamination from the adhesive.
8. Prepare the adhesive. Dispense a small amount of metal bonding adhesive from the cartridge to make sure of an even flow of both components. Attach the mixing tip and dispense a mixing tip length of adhesive to make sure of correct mix ratio.
9. **NOTE:** *Refer to product label for cure times and handling instructions.*

Apply a 6-9 mm (0.23-0.35 in) bead of adhesive to the vehicle prepared flange surface. Remove the tape from the plug weld areas.

10. Place the service part(s) in the correct position on the vehicle.
 - When positioned, do not pull the component away from the vehicle. If repositioning is necessary, slide the service part(s). This will make sure of correct contact between the components and adhesive.
11. Clamp evenly and tightly. The adhesive contains glass beads which will prevent over-clamping the component.
12. **NOTE:** *Welding can be carried out anytime during the adhesive curing process or after the adhesive is fully cured.*

NOTE: *If welding will not be carried out immediately, refer to product label for cure times and handling instructions. Clamps may be removed immediately after the component is welded.*

Wipe excess adhesive from the panel before it cures.

13. Finish any cosmetic section seams with fiber-filled body filler. Rough sand the filler, apply conventional body filler after the adhesive cures and block-sand the area.
14. Use seam sealer wherever a cosmetic seam sealer is required.
15. Mix and apply primer surfacer per Ford-approved paint recommendations.
16. Mix and apply basecoat per Ford-approved paint recommendations.

17. Mix and apply clearcoat per Ford-approved paint recommendations. Refinishing materials may be force-dried following paint manufacturer's recommendations.
18. Apply anti-corrosion treatment to the repair area as required. For additional information, refer to Restoring Corrosion Protection Following Repair in this section.